

|                                                         |                                       |                                               |                                                         |                                           |                                           |                                               |
|---------------------------------------------------------|---------------------------------------|-----------------------------------------------|---------------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                         | $\mathcal{K}_{0^+}^{\#1}$             | $\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                                         |                                           |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1}{}^\dagger$                     | 0                                     | 0                                             |                                                         |                                           |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1}{}^{\alpha\beta\chi}{}^\dagger$ | 0                                     | 0                                             | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$               | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $-k^2 \frac{(4)}{\kappa_3}$           | $\frac{k^2 (4)}{2 \sqrt{2} \kappa_4}$         | 0                                                       | 0                                         |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $\frac{k^2 (4)}{2 \sqrt{2} \kappa_4}$ | $\frac{\Upsilon_1 k^2}{2}$                    | 0                                                       | 0                                         |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$          | 0                                     | 0                                             | $k^2 \frac{(4)}{\kappa_1}$                              | $-\frac{k^2 (4)}{\sqrt{2} \kappa_1}$      |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$          | 0                                     | 0                                             | $-\frac{k^2 (4)}{\sqrt{2} \kappa_1}$                    | $\frac{k^2 (4)}{2 \kappa_1}$              | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                         |                                       |                                               | $\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | 0                                         | 0                                         |                                               |
|                                                         |                                       |                                               | $\mathcal{K}_{2^+}^{\#1}{}^{\alpha\beta\chi}{}^\dagger$ | 0                                         | 0                                         |                                               |

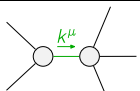
|                                                         |                                                        |                                                        |                                                         |                                           |                                           |                                               |
|---------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                         | $\mathcal{J}_{0^+}^{\#1}$                              | $\mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi}$          |                                                         |                                           |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1}{}^\dagger$                     | 0                                                      | 0                                                      |                                                         |                                           |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1}{}^{\alpha\beta\chi}{}^\dagger$ | 0                                                      | 0                                                      | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$               | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$ | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{\Upsilon_1 k^2}{2 \text{Det}(1^+)}$             | $-\frac{k^2 (4)}{2 \sqrt{2} \text{Det}(1^+) \kappa_4}$ | 0                                                       | 0                                         |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $-\frac{k^2 (4)}{2 \sqrt{2} \text{Det}(1^+) \kappa_4}$ | $-\frac{k^2 (4)}{\text{Det}(1^+) \kappa_3}$            | 0                                                       | 0                                         |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$          | 0                                                      | 0                                                      | $\frac{4}{9 k^2 (4) \kappa_1}$                          | $-\frac{2 \sqrt{2}}{9 k^2 (4) \kappa_1}$  |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$          | 0                                                      | 0                                                      | $-\frac{2 \sqrt{2}}{9 k^2 (4) \kappa_1}$                | $\frac{2}{9 k^2 (4) \kappa_1}$            | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                         |                                                        |                                                        | $\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | 0                                         | 0                                         |                                               |
|                                                         |                                                        |                                                        | $\mathcal{J}_{2^+}^{\#1}{}^{\alpha\beta\chi}{}^\dagger$ | 0                                         | 0                                         |                                               |

Abbreviations used in matrices

$$\Upsilon_1 == \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4} \&\& \text{Det}(1^+) == -\frac{1}{8} k^4 (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2$$

Lagrangian

$$\begin{aligned} & \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}^\delta{}_{\chi\delta} \partial^\chi \mathcal{K}^\alpha{}_{\alpha}{}^\beta - \frac{(4)}{\kappa_1} \partial_\chi \mathcal{K}^\delta{}_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_{\alpha}{}^\beta + \frac{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\alpha\chi} + \\ & \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} - \frac{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha} + \frac{1}{2} \frac{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} + \frac{1}{2} \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} \end{aligned}$$

|                                                                                    |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                             |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                             |
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |
| $\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} == 0$                                      | 1                                                             | $\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$<br>$\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             |
| $\mathcal{J}_{0^+}^{\#1} == 0$                                                     | 1                                                             | $\partial_\beta \mathcal{J}^\alpha{}_{\alpha}{}^\beta == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             |
| $\mathcal{J}_{1^+}^{\#1\alpha} + 2 \mathcal{J}_{1^+}^{\#2\alpha} == 0$             | 3                                                             | $\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_{\beta}{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^\beta{}_{\beta}{}^\alpha == 3 \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$                                      | 5                                                             | $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$<br>$4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha}{}_{\delta} ==$<br>$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$<br>$2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta}{}_{\delta}$ |                             |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$                                          | 5                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^\chi{}_{\chi}{}^\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\chi{}_{\chi}{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                             |
| Total # constraint(s):                                                             | 15                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                             |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Residue                     |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $-\frac{1}{\kappa_1^{(4)}}$ |
| Resolved unitarity condition(s):                                                   | $\frac{(4)}{\kappa_1} < 0$                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                             |